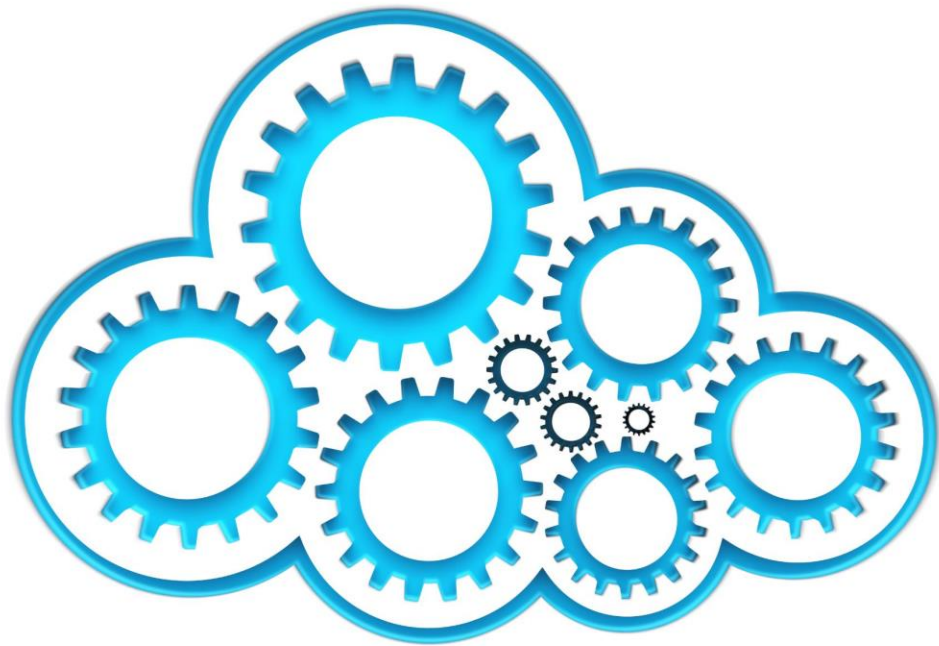


Azure App Registration Secret Rotation Script

Automating secure credential updates for cloud applications

Introduction

Overview of the Script



Secret and Certificate Rotation

The script automates secure rotation of secrets and certificates for Azure Active Directory App Registrations.

Secure Storage and Access Management

Expired credentials are securely stored in Azure Key Vault with managed identity managing access policies.

Network and Policy Configuration

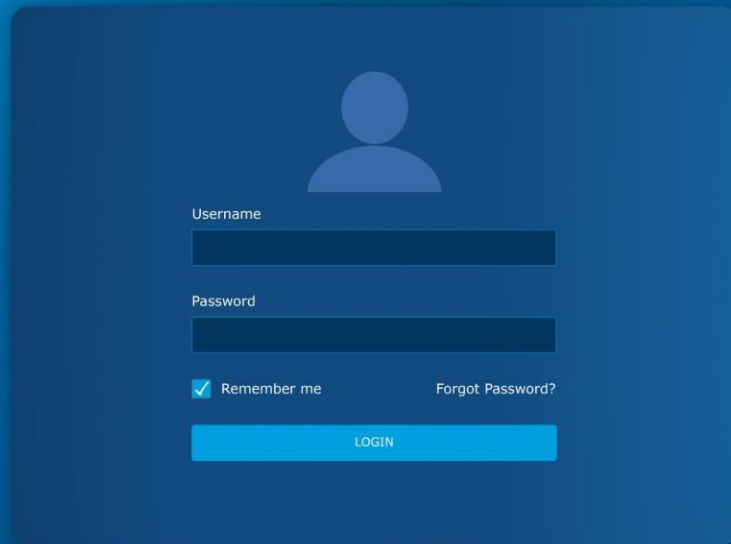
The script manages network firewall rules and access policies to enable secure secret management.

Error Handling and Logging

Robust error handling and logging ensure operational efficiency and security monitoring.

Prerequisites

Required Modules and Permissions



Module Installation

Install Az and Microsoft.Graph.Applications modules using PowerShell Import-Module commands for script execution.

Managed Identity Requirement

Script uses managed identity for authentication, requiring a system-assigned identity in Azure Automation or VM.

Access and Permissions

Managed identity needs Contributor or Owner subscription access and Key Vault Administrator permissions to manage secrets.

Microsoft Graph API Permissions

Permissions like Application.ReadWrite.All and Directory.ReadWrite.All are required to interact with Microsoft Graph API securely.

Environment Setup



Configuration Parameters

Essential Configuration Parameters

Subscription name, Key Vault name, resource group, and App Registration display name are required for the script.

Purpose of Parameters

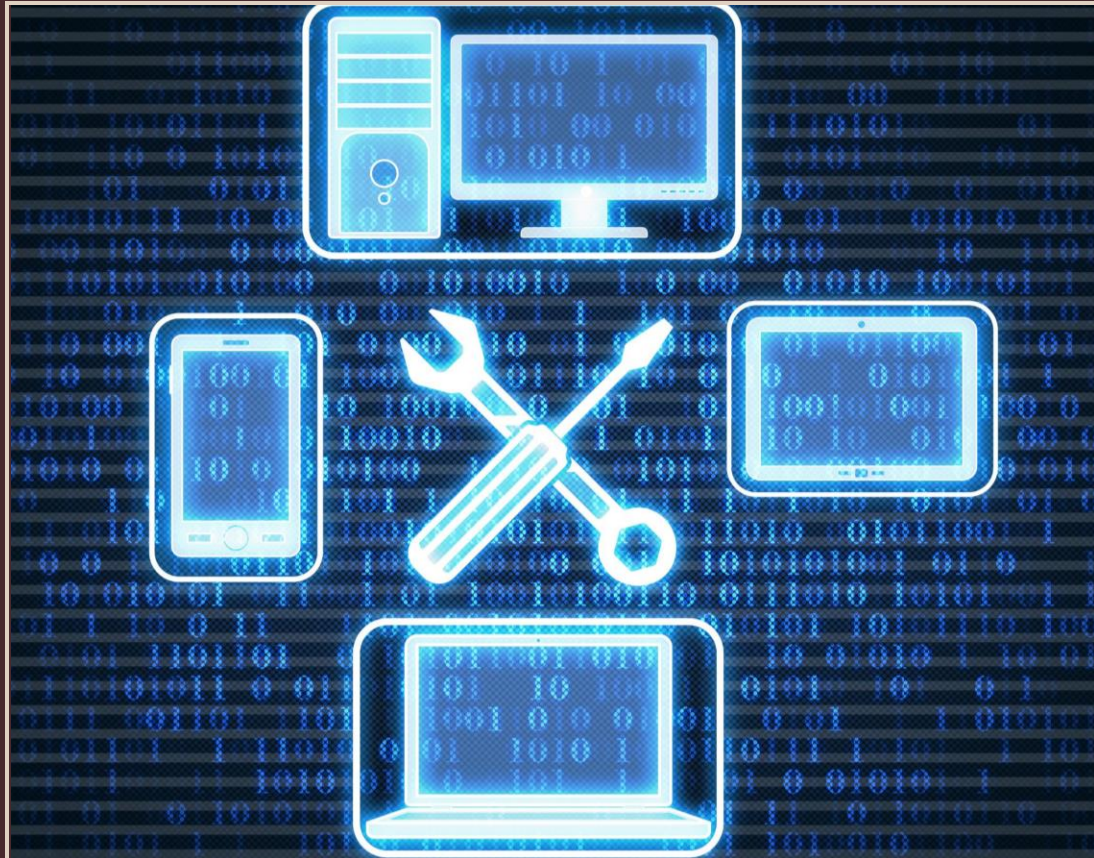
These parameters help identify and manage Azure resources accurately within the script.

Secure Operations

Proper configuration ensures authentication, resource access, and secure secret rotation and storage.

Script Breakdown

Functional Components



Script Initialization

The script starts with module imports and authenticates using managed identity for secure access.

Secret Rotation Function

`rotate_expired_secrets` updates firewall rules, creates new secrets, and stores them securely.

Expired Secret Removal

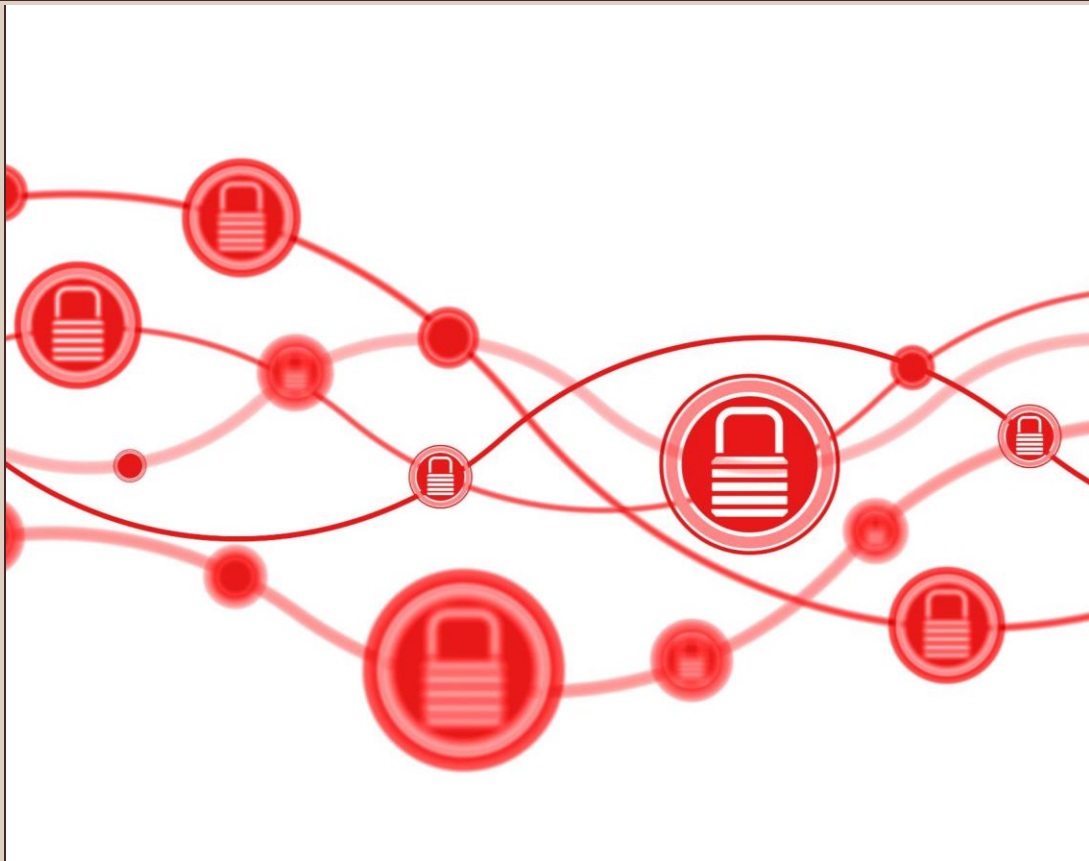
`remove_expired_credential_secrets` deletes expired credentials from App Registrations for security.

Expiration Audit and Execution

Audit flags credentials expiring soon and rotation loop processes secret replacement systematically.

Key Features

Security and Automation Enhancements



Dynamic Network Access Control

The script dynamically updates Key Vault network ACLs with IP addresses to maintain secure access during secret operations.

Error Handling and Reliability

Error handling mechanisms capture IP-related access errors and retry operations to improve script reliability.

Credential Lifecycle Management

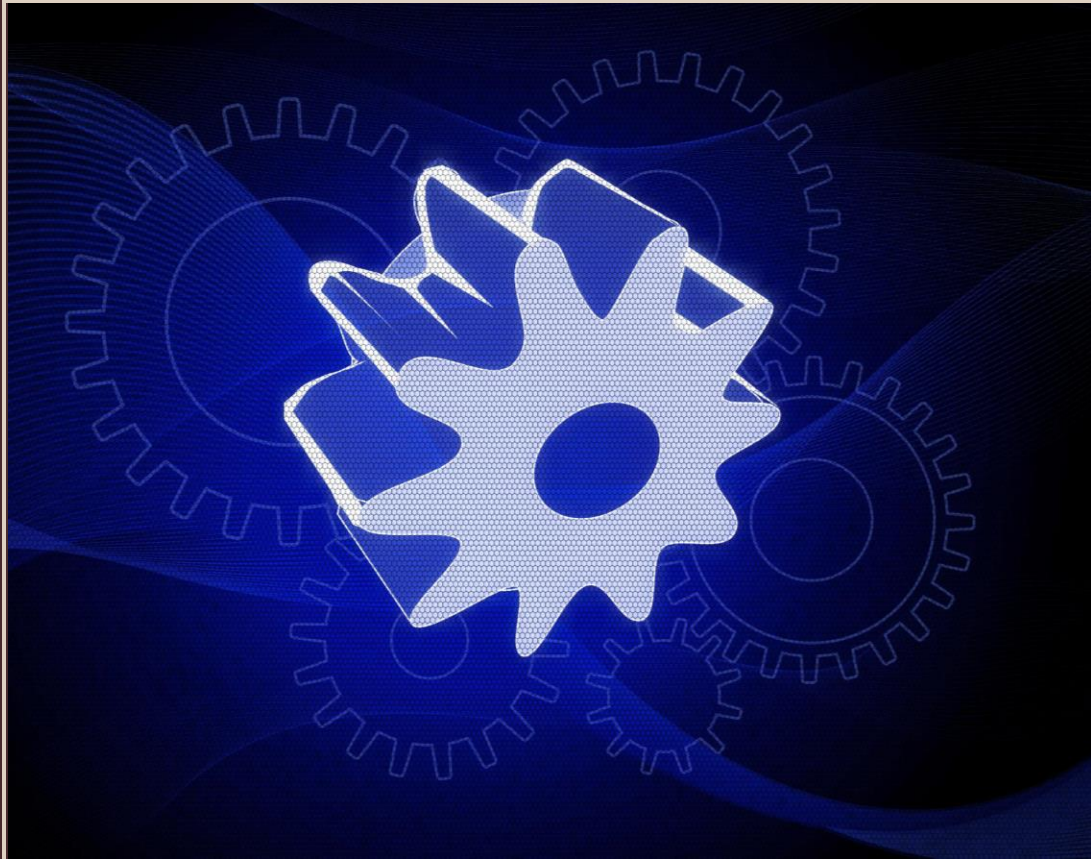
Secrets are securely rotated and stored to reduce risks from expired or compromised credentials.

Comprehensive Logging and Traceability

Logging ensures traceability of IP additions and errors, supporting auditing and compliance requirements.

Demo Results

Execution Output Screenshot



Secret Expiration Audit

The script audits secret expiration dates, highlighting expired credentials in red and active ones in green for clarity.

Secret Rotation Process

Expired secrets are removed and replaced with new credentials, ensuring continuous security and access management.

Secure Storage in Azure Key Vault

New credentials are securely stored in Azure Key Vault, enhancing security and centralized secret management.

Automation Validation

Output confirms successful script execution and effective automation, providing visual feedback and validation.

Execution Output Screenshot

```
434
435
436
437
438 }
439
440 # Get all App Registrations
441 $appRegistrations = Get-AzADApplication
442
443 foreach ($app in $appRegistrations) {
444     # Check Password Credentials
445     foreach ($cred in $app.PasswordCredentials) {
446         $daysRemaining = ($cred.EndDateTime - (Get-Date)).Days
447         Write-Host "App: $($app.DisplayName) - Secret expires on:" -NoNewline
448
449         if ($daysRemaining -le $thresholdDays) {
450             Write-Host " $($cred.EndDateTime)" -ForegroundColor Red -BackgroundColor White
451             $seccertstatus = "Expired"
452         } else {
453             Write-Host " $($cred.EndDateTime)" -ForegroundColor Green
454             $seccertstatus = "Active"
455         }
456     }
457 }
458
459 }
460
461
```

```
Write-Host " $($cred.EndDateTime)" -ForegroundColor Green
$seccertstatus = "Active"
}
}
}

App: wolffentpOAI - Secret expires on: 09/11/2025 23:12:57
App: wolffautomationsp - Secret expires on: 09/11/2025 23:15:09
App: wolffentraadmins - Secret expires on: 09/11/2025 23:17:27
App: aro-j8k6x4vn - Secret expires on: 09/11/2025 23:19:41
App: wolffentpweb - Secret expires on: 09/11/2025 23:21:54
App: wolffb2cblock - Secret expires on: 09/10/2026 15:34:28
App: arcsvcprinmi - Secret expires on: 02/05/2026 16:14:52
App: wolffentpwebsite - Secret expires on: 05/14/2027 23:39:43
App: aro-tcn6i4e4 - Secret expires on: 12/05/2026 20:34:25
```

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```
missions to (Key Vault Managed) Storage :
Tenant ID : e594
Object ID : de79
Application ID :
Display Name :
Permissions to Keys :
Permissions to Secrets : set,
Permissions to Certificates :
Permissions to (Key Vault Managed) Storage :

Microsoft.Azure.Commands.KeyVault.Models.PSKeyVault
Fault Action : All
Pass : Azure
Rules : 71.2
Virtual Network Rules :

Microsoft.Azure.Management.KeyVault.Models.Vault
Subscriptions/5755893a-8056-4ba8-9916-1133c80a80f3
Wolffkv
string
Vault
SequenceStart, Team, Purpose, Email...}

Name Value
=====
SequenceStart 1
Team CSU
Purpose automation and spn use
Email jerrywolff@microsoft.com
BusinessUnit West OU
SequenceStop 1
Owner Jerry wolff

"Update-AzADApplication_UpdateExpanded" on target
"Remove-AzADApplicationPassword_RemoveExpanded"
"App Registration for wolffautomation"
Placement and stored in wolffkv - wolffautomation

Automation
10-1ec9-4192-a8d4-a9111f8cfa7
a8-9916-1133c80a80f3 MSI050342 AzureCloud

Press Ctrl-B to break into debugger.
```

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```
711c-41d1-b774-bf08f57a328a :
on ID :
ame :
ns to Keys :
ns to Secrets :
list :
ns to Certificates :
ns to (Key Vault Managed) Storage :
. Azure.Commands.KeyVault.Models.PSKeyVaultRuleSet
ction :
ices :
$.52/32 :
etwork Rules :
. Azure.Management.KeyVault.Models.Vault
tions/5755893a-8056-4ba8-9916-1133c80a8
rceGroups/Adminrg/providers/Microsoft.K
aults/wolffkv
start, Team, Purpose, Email...}
==== Value
start 1
CSU
automation and spn use
Jerrywolff@microsoft.com
nit West OU
top 1
Jerry wolff
9/10/2030 23:30:08
on: 09/10/2030 23:32:22
on: 09/10/2030 23:34:41
9/10/2030 23:36:53
9/10/2030 23:39:08
09/10/2026 15:34:28
2/05/2026 16:14:52
nt: 05/14/2027 23:39:43
2/05/2026 20:34:25
res on: 12/04/2026 20:49:47
expires on: 07/18/2027 00:55:53
: 04/22/2026 20:38:33
```

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